(1) P is a stocastic matrix, with $\underline{n \times n}$.

$$\sum_{j=1}^{n} P_{ij} = 1, \ \forall i$$

Show that $P^T \lambda = \lambda$, $\lambda \geq 0$ has non-zero solution.

Farkka lemma state Either (1) (or) (2) of the following must Hold

(1) $\exists x \neq 0$, s.t. $Ax = 0$, $x \geq 0$

(2) $\exists y$ s.t. $y^T A > 0$

We show that (2) is not Possible. Hence (1)

$$P^T \lambda = \lambda$$

$$P^T \lambda - \lambda = 0 \Rightarrow \boxed{(P^T - I)\lambda = 0} \quad (i)$$

Let $A = P^T - I$ Here.

Let us show that

$Ay > 0$ is not possible

i.e $(P^T - I)\lambda > 0$ is not Possible

Let's <u>show</u>: By contradiction (as following

(1) Let's assume $(P^T - I)\lambda > 0 \Rightarrow P^T\lambda > \lambda$

In Frakka-form let $(P^T - I) = A$

$$A\lambda > 0 \Rightarrow$$

we are given $\sum_{j=1}^{n} P_{ij} = 1, \forall i$ (sum of probabilities)

Assume λ is normalized. $P^T\lambda$ is weighted average of probabilities (similar to that)

$\boxed{\text{Let's Assume } A\lambda > 0 \Rightarrow P^T\lambda > \lambda}$

Given $\sum_{j=1}^{n} P_{ij} = 1$, the product of probability row, against normalized λ, can not be greater than the max. λ in the corresponding row

A simple e.g illustrate

$$\begin{pmatrix} 0.6 & 0.5 \\ 0.2 & 0.8 \end{pmatrix} \begin{pmatrix} .2 \\ 0.8 \end{pmatrix} = \begin{pmatrix} 0.4 \\ 0.68 \end{pmatrix} \qquad 0.68 < 0.8$$

mathematically

every row of $(P\lambda) \Rightarrow (P\lambda)_i = \sum_{j=1}^{n} P_{ij}\lambda_j$

If (i^*) is the row index corresponding to $\max \lambda$, denoted as λ_{i^*}

$$\Rightarrow \left((P\lambda)_{i^*} = \sum_{j=1}^{n} P_{i^*j}\lambda_j \right) \leq \lambda_{i^*}$$

where λ_{i^*} is the $\max \lambda_i$

$(P\lambda)_{i^*} \leq \lambda_{i^*}$ is a contradiction, given our assumption we assumed $P^T\lambda > \lambda_{i^*}$

Hence $\exists\, \lambda \neq 0$, s.t $(P^T - I)\lambda = 0, \lambda \geq 0$

$\Rightarrow P\lambda = \lambda, \lambda \geq 0$

(2)

Per Farkkai lemma —

$\exists x \neq 0$, s.t. $Ax = 0$, $x \geq 0$.

2a

Here we are given $P \in R^m$, $A^T P = 0$, $P \geq 0$ and $\sum_{i=1}^{m} P_i = 1$. (implies $P \neq 0$).

$$\min \max_{i=1 \text{ to } m} (a_i^T x - b_i) \; , \; x \in R^n$$

~~P is Probability (or can be taken to be or $\sum_{i=1}^{m} P_i = 1$).~~

Let $v = \min \max_{i=1 \text{ to } m} (a_i^T x - b_i)$

As discussed in HW, Problem (1) and ~~from~~ seen when we take $P_i (a_i^T x - b_i)$

For $i = 1$ to m $\sum_{i=1}^{m} P_i (a_i^T x - b_i) \leq \max_i (a_i^T x - b_i)$

(This was discussed in HW 1) using that result.

$$\sum_{i=1}^{m} P_i (a_i^T x - b_i) = \sum_{i=1}^{m} P_i a_i^T x - \sum_{i=1}^{m} P_i b_i$$

we are given $A^T P = 0$ we can see that $\sum_{i=0}^{m} P_i a_i^T x = P^T A x = 0$

~~Hence~~ $\sum_{i=1}^{m} p_i (a_i^T x - b_i) = 0 - \sum_{i=1}^{m} p_i b_i$

$$= -p^T b$$

$$\sum_{i=1}^{m} p_i (a_i^T x - b_i) \leq \max_i (a_i^T x - b_i)$$

$$\boxed{\max_i (a_i^T x - b_i) \geq -p^T b}$$

$$\boxed{v \geq -p^T b} \Rightarrow -p^T b \leq v$$

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(25) ~~min max~~ $\cancel{(a_i^T x - b_i)}$, ~~$x \in R^n$~~,

25 $$\min_{x} \left[\max_{i=1 \text{ to } m} (a_i^T x - b_i) \right]$$

Let's state this as LP:- Let

$$\max_{i=1 \text{ to } m} (a_i^T x - b_i) \text{ be denoted as } Z$$

~~As~~ Stating it as LP:-

$$\min \; Z$$

$$\text{s.t} \quad a_i^T x - b_i \leq Z \quad (\text{min max is upper bound})$$

$$\Rightarrow \min \max \; Z + P^T(a_i^T x - b_i - Z_i)$$

Let's introduce P_i, s.t., $\sum_{i=1}^{m} P_i = 1$, $P^T e = 1$

$$\Rightarrow \min \left[\max_{P \geq 0} \left(Z_i + P^T a_i^T x - P_i b_i - P_i Z_i \right) \right]$$

$$\Rightarrow \max_{P \geq 0} \min_{x_i, Z_i} -P_i b_i + \underbrace{(1 - P_i)}_{P^T e = 1} Z_i + P_i^T a_i^T x$$

writing it in matrix form

$$\Rightarrow \max_{P \geq 0} \; -P_i b_i$$

$$\text{s.t} \quad P_i = 1$$

$$P_i a_i = 0$$

$$P \geq 0$$

writing it in matrix form

$$\therefore \left[\Rightarrow \max_{P \geq 0} \; -Pb \quad \text{s.t} \quad A^T P = 0, \; P^T e = 1 \text{ and } P \geq 0 \right]$$

2(b) In 2(a) we found best possible upper bound $\boxed{-p^T b} \leq V$.

~~The dual of the Primal has the same optimal solution because of the theorem of Strong duality~~

2b By LP weak duality $\Rightarrow$

$$(\max \min) \leq (\min \max)$$ (defined in 2a)

Hence

$$-p^T b \leq V$$

(3) $\min c^T x$, s.t. $Ax \geq c$, $x \geq 0$

Prove that if x^* satisfies $Ax^* = c$, $x^* \geq 0$ then x^* is optimal, (A is symmetric)

We are given $Ax^* = c$, $x^* \geq 0$

$\forall x \quad Ax \geq c$, $x \geq 0$

multiply ~~c^T bt~~ by $(x^*)^T$ on Both sides

$$(x^*)^T A x \geq (x^*)^T c$$

A is symmetric matrix $\therefore (x^*)^T A x = x^T A (x^*)^T$

$\Rightarrow$ ~~$x^T A (x^*)^T \geq c^T x^*$~~ $[\because (AB)^T = B^T A^T]$

$\Rightarrow x^T A x^* \geq c^T x^*$ $[\because (AB)^T = B^T A^T]$

Since $Ax^* = c$

$\Rightarrow x^T c \geq c^T x^*$

$c^T x \geq c^T x^*$

Any $c^T x$ is greater than or equal to $c^T x^*$

Implies $c^T x^*$ is minimum

Hence x^* is the optimum solution

(4) consider $\min: c_1 a_1 + c_2 a_2 + c_3 a_3$

$$\text{s.t. } Ax \geq b \; ; \; \forall b \in \mathcal{U}$$

uncertainty set $\mathcal{U}$ is unit box in "m" dimension,

i.e. $\mathcal{U} = \{ b^{\circ} : 0 \leq b_i \leq 1, i = 1 \ldots m, \sum_{i=1}^{m} b_i \leq 1 \}$

S.T. This is Equivalent to the following LP:-

$$\min \; c_1 a_1 + c_2 a_2 + c_3 a_3$$

$$\text{s.t. } Ax \geq \hat{b}$$

where $\hat{b} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$

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① Robust LP is LP where we do not exactly know what the Parameters of our LP are

② we only know approximately the parameters, but not Exactly

we have uncertainty in b, call it $\mathcal{U}_b$

③ In this minimization problem we should minimize for worst case

Here we have Type-3 uncertainty on the RHS of linear constraint.

(4) with uncertainty we solve the LP for absolute worst case.

The robust constraints (TYPE-3, on R-1)

$$a_i^T x_i \geq b_i, \quad \forall b_i \in \mu$$

$$0 \leq b_i \leq 1, \quad \sum_{i=1}^{m} b_i = 1$$

The worst case, b_i, i.e. the LARGEST b_i

$i = 1$ to m $\quad b_i \leq 1$, worst case $b_i = 1$

$i = 1$ to m, $b_i \leq 1$, but WORST CASE $b_i = 1$

$\sum_{i=1}^{m} b_i \leq 1$, mean not all b_i can be 1 at the same time

$i = 1$ to m, worst-case $b_j = 1$, $\Rightarrow i \neq j$, $b_i = \emptyset$

However worst case $b_i = 1$

Hence $Ax \geq \hat{b}$, $\hat{b} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$ is the ~~robust constraint~~ is the robust constraint.

Hence the equivalent LP is

min : $c_1 x_1 + c_2 x_2 + c_3 x_3$, s.t. $Ax \geq \hat{b}; \hat{b}^2 \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$